

Authors' Responses

Journal Title:	IEEE Robotics and Automation Letters
Reference Number:	26-3841v1
Title of Paper:	VLM-DexSafeDagger: VLM-Guided Intervention Policy for Safety-Critical Visuomotor Skill Distillation
Authors:	Anonymous Authors

We sincerely thank the Editor and the reviewers for their careful evaluation of our manuscript and for their constructive comments. Their feedback has helped us identify where the contribution, safety claims, experimental evidence, and implementation details require clearer presentation.

We are preparing a major revision of the manuscript. In the revised version, we will:

- 1) clarify the specific contribution of VLM-guided threshold adaptation through comparisons with conventional fixed and adaptive alternatives;
- 2) revise the safety terminology and explicitly delimit the empirical, task-specific scope of the reported safety results;
- 3) strengthen the evaluation of the Bellman critic, teacher policies, real-world experiments, and resource usage; and
- 4) provide the implementation details needed to reproduce the VLM-based threshold tuner and the intervention pipeline.

Detailed responses to each comment are provided below. Reviewer comments are reproduced in black, and our responses are shown in blue. Red bracketed text marks details that will be completed after the corresponding revision or experiment has been finalized. Changes in the revised manuscript will be highlighted in blue.

A Appendix Materials

A.1 RL Teacher Training and Safety Details

Table 1 contrasts the actor’s noisy, deployable signals with the centralized critic’s clean, privileged simulator state. The palm and five fingertips constitute the six hand links. A “twist” is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	87	133	

“–” denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

Termination conditions. [Report the numerical trigger, units, measured signal, and physical justification for harmful collision, object escape, palm flipping, and the hand-too-far/workspace condition. State how simultaneous triggers are resolved and whether the same definitions are used for teacher training, distillation, and evaluation. Report the available contact-force or other severity signal and define the severity-weighted safety metric.] Requested by Reviewer #22.

Runs and random seeds. Teacher performance in Main Paper Table II is evaluated over 200 episodes per object, and the real-world results are evaluated over 30 episodes per object. Each of the 40 object-specific teacher policies is trained with five independent random seeds $\{1, 2, 3, 4, 5\}$, with the same evaluation protocol applied to every seed. For each seed, teacher performance is evaluated on all 40 objects. The table below reports the mean $\pm$ standard deviation across the 40 object-level rates for lift success, overall unsafe events, and the four unsafe-event categories separately for each seed. Requested by Reviewers #14 and #22.

Table 2: Teacher policy performance across all 40 objects for each random seed (200 evaluation episodes per object and seed). Each entry is the mean $\pm$ standard deviation across the 40 object-level rates. Numerical entries are provisional synthetic placeholders pending verification with the measured evaluation data.

Metric (%)	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
Lift success	86.7 ± 7.8	87.4 ± 7.5	86.9 ± 8.0	87.8 ± 7.3	87.1 ± 7.7
Unsafe rate avg.	14.2 ± 6.1	13.6 ± 5.8	14.0 ± 6.3	13.2 ± 5.6	13.8 ± 6.0
Harmful collision	3.1 ± 3.5	2.9 ± 3.3	3.2 ± 3.6	2.8 ± 3.2	3.0 ± 3.4
Object escape	9.0 ± 5.2	8.8 ± 5.0	8.9 ± 5.3	8.7 ± 4.9	8.9 ± 5.1
Palm flipped	1.2 ± 2.4	1.1 ± 2.3	1.1 ± 2.3	1.0 ± 2.2	1.1 ± 2.3
Hand too far	0.9 ± 1.9	0.8 ± 1.8	0.8 ± 1.8	0.7 ± 1.7	0.8 ± 1.8

Teacher safety and disagreement handling. Main Paper Table II reports representative teacher unsafe-event rates. Across all 40 teacher policies, unsafe-event rates range from 5–30%. Because an intervention transfers control to an imperfect teacher, the framework’s safety performance is restricted by the teacher’s own performance ceiling. Teacher takeover

can reduce unsafe exploration, but it does not provide a formal safety guarantee. **[Define how the controller handles disagreement between the teacher action and the critic’s assessment, including the fallback used when neither candidate action satisfies the acceptance rule.]** *Requested by Reviewer #22.*

A.2 VLM Prompt and Inference Details

The VLM prompt specifies the system and user requirements, intervention rule, multimodal input format, task description, and output keys, as shown below.

VLM Prompt Example

System:

You are a robotics safety assistant. Your task is to assess the current safety-autonomy operating point of a visuomotor policy.

Inputs:

- Current visual observation: <IMAGE>
- Teacher-student disagreement: {d}
- Predicted success probability: {q_succ}
- Intervention rate: {r_int}
- Unsafe-event rate: {r_unsafe}

Task:

Determine whether the current intervention boundary should become more conservative or more permissive.

Return:

```
{
  "disagreement_threshold": ...,
  "success_threshold": ...,
  "reason": ... }
```

Requested by Reviewers #14 and #22.

Model and input configuration. We used the OpenAI GPT-5.5 multimodal model through the Responses API (model="gpt-5.5") with reasoning effort set to high (reasoning.effort="high"). The model received both textual rollout statistics and RGB images. All other generation parameters were left at their API defaults. We issued the first VLM query after 2,000 episodes and subsequently queried it every 10,000 distillation episodes, resulting in 10 VLM calls over the full 100,000-episode training run and approximately 70,000 tokens of total usage per run. *Requested by Reviewers #14 and #22.*

Output processing and threshold updates. **[Provide the machine-readable output schema, parsing and validation logic, permitted threshold ranges, clipping and smoothing rules, and query/update frequency. Explain exactly how the parsed output changes the intervention policy. State explicitly whether VLM adaptation is used only during simulation distillation, during real-world student training, or during physical deployment.]** *Requested by Reviewers #14 and #22.*

Reliability and resource usage. **[Report retry and timeout behavior, malformed-output handling, deterministic fallback, measured latency, token usage, number of calls, and monetary cost per training run.]** *Requested by Reviewer #22.*

A.3 Additional Experimental Results

Bellman critic evaluation. [State how safety risk and task success are separated, or justify the retained target. Report AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and the false-negative rate for unsafe episodes at the deployed threshold.] *Requested by Reviewers #14 and #22.*

VLM-necessity and threshold-adaptation ablations. [Report matched comparisons for fixed thresholds, grid-search and Bayesian-optimization selection, rule-based adaptation, curriculum scheduling, a small learned threshold controller, an LLM receiving rollout statistics only, the VLM receiving images only, the VLM receiving statistics and images, and a visual-input removal or mismatch control. Include task success, unsafe-event rate, and intervention cost under matched budgets.] *Requested by Reviewers #14 and #22.*

Training and resource comparison. [For Vanilla DAGger, SafeDAGger, critic-based intervention, and VLM-guided intervention, report environment steps, student updates, teacher queries or labels, teacher-controlled steps, intervention rate, compute time, VLM calls and cost, final success and unsafe-event rates, and results over multiple seeds.] *Requested by Reviewer #22.*

Complete real-world evaluation. We will report results for the remaining 16 real-world test objects that are not shown in the manuscript’s six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes. Object names and numerical results are retained as placeholders pending the expanded evaluation. *Requested by Reviewers #14 and #22.*

Table 3: Template for per-object experimental results for the remaining 16 objects. Object names and numerical entries are placeholders.

Metric (%)	Object 7	Object 8	Object 9	Object 10	Object 11	Object 12	Object 13	Object 14
Lift Success	–	–	–	–	–	–	–	–
Unsafe Rate Avg	–	–	–	–	–	–	–	–
Harmful collision	–	–	–	–	–	–	–	–
Object escape	–	–	–	–	–	–	–	–
Palm flipped	–	–	–	–	–	–	–	–
Hand too far	–	–	–	–	–	–	–	–
Metric (%)	Object 15	Object 16	Object 17	Object 18	Object 19	Object 20	Object 21	Object 22
Lift Success	–	–	–	–	–	–	–	–
Unsafe Rate Avg	–	–	–	–	–	–	–	–
Harmful collision	–	–	–	–	–	–	–	–
Object escape	–	–	–	–	–	–	–	–
Palm flipped	–	–	–	–	–	–	–	–
Hand too far	–	–	–	–	–	–	–	–

[Replace the object labels and dashes with the remaining 16 per-object results. State 30 episodes per object, report 95% confidence intervals and statistical comparisons, and provide aggregate results separately for seen and unseen objects across the full 22-object evaluation.] *Requested by Reviewers #14 and #22.*

Physical baseline comparison. [Report the feasible matched real-robot baseline using the same objects, initial-condition protocol, and number of trials, including lift success, unsafe-event incidence, uncertainty, and the statistical comparison to VLM-DexSafeDAGger.] *Requested by Reviewer #14.*

B Response to the Associate Editor

We thank the Associate Editor for summarizing the central issues raised during review. We address each issue below and provide detailed point-by-point responses to the reviewers in the following sections.

Comment 1: The experiments do not sufficiently separate the contribution of the VLM from the effect of adaptive threshold tuning. Additional comparisons and ablations are needed to clarify whether the VLM itself provides a benefit.

[Response:] We agree that the comparison between a fixed intervention policy and VLM-guided adaptation does not, by itself, isolate the source of the improvement. In the revision, we will add controlled comparisons covering fixed threshold selection, conventional adaptive threshold rules, a learned threshold controller, statistics-only language-model input, and statistics-plus-image VLM input. All methods will use the same teacher, student, critic, training budget, and evaluation protocol. We will report both task performance and intervention cost so that safety improvements cannot be attributed only to increased teacher control. **[Insert the finalized methods, principal quantitative result, and revised table/figure locations.]**

Comment 2: The framework provides empirical intervention rather than a formal safety guarantee, and unsafe episodes remain in the real-world experiments. The scope of the “safety-critical” claim should therefore be clarified and better supported.

[Response:] We agree. The revised manuscript will explicitly state that the framework does not guarantee constraint satisfaction and that safety violations remain possible. We will replace overly strong uses of “safety-critical” with terms such as “safety-aware” or “empirically risk-reducing,” and we will limit our conclusions to the task-specific unsafe events measured in the experiments. **[List the affected sections and the final revised title, if applicable.]**

Comment 3: The Bellman success-value critic, VLM reproducibility, number of real-world trials, baseline coverage, and reporting for all tested objects need to be strengthened.

[Response:] We agree and will expand both the evaluation and the implementation description. The revision will include critic error and calibration analyses, the complete VLM prompt and output-processing procedure, the number of random seeds, matched baseline comparisons, uncertainty estimates, and aggregate as well as per-object real-world results. **[Insert the completed analyses and their manuscript or supplementary-material locations.]**

C Response to Reviewer #14

We sincerely thank the reviewer for the careful assessment of our work and for the constructive suggestions. Below are our point-by-point responses.

Comment 1: The current comparison between DexSafeDagger and VLM-DexSafeDagger compares fixed thresholds with VLM-adapted thresholds and therefore does not show whether the improvement comes from threshold adaptation, language reasoning, or visual information. Please compare cases with and without an LLM/VLM and with and without statistics and/or images.

[Response:] We agree. We will add controlled ablations that separate adaptation, language reasoning, and visual input. In particular, the revision will compare a fixed operating point, a non-language adaptive controller, an LLM receiving rollout statistics only, a VLM receiving rollout statistics and selected images, and a visual-control condition in which the image information is removed or mismatched. These variants will be evaluated under matched training and intervention budgets.

[Insert final ablation definitions, results with uncertainty, and table/figure references.]

Comment 2: The related work on adaptive intervention should be expanded. In particular, AIM, EnsembleDagger, and HG-Dagger are relevant. Since adaptive intervention can be achieved without a VLM, please clarify how VLM-DexSafeDagger differs from these methods and whether the VLM provides an advantage over simpler adaptive approaches.

[Response:] We agree and will expand the related-work discussion to include AIM, EnsembleDagger, and HG-Dagger. We will distinguish uncertainty-based gating, human-gated intervention, learned intervention proxies, and our setting of autonomous multi-object visuomotor distillation with a two-channel disagreement/value gate. We will also ground the claimed benefit of the VLM in the new adaptive-baseline experiments rather than in a conceptual distinction alone. **[Insert citations and the revised section location.]**

Comment 3: Please provide the VLM prompt and explain how images are selected and how the VLM outputs are used. The Bellman success-value critic should also be evaluated, especially for errors on unsafe cases. Please clarify the number of runs or random seeds and the scope of the term “safety-critical.”

[Response:] We agree. The supplementary material will provide the exact prompt, model identifier, inference settings, image-selection rule, output format, parser, threshold clipping and smoothing, query schedule, retry behavior, and deterministic fallback. We will also report critic discrimination and calibration metrics, with particular attention to false negatives on unsafe episodes, and will state the number of independent runs used for every result. Finally, we will revise the safety terminology as described in our response to the Associate Editor. **[Insert model version, number of seeds, critic metrics, and section/appendix references.]**

Comment 4: Only VLM-DexSafeDagger is evaluated on the real robot. This does not sufficiently demonstrate the proposed concept; a physical baseline comparison is needed.

[Response:] We agree that a matched physical comparison would provide stronger evidence. We will evaluate the distilled policy against **[specify the feasible real-world baseline or baselines]** using the same object set, initial-condition protocol, and number of trials. We will report lift success, unsafe-event incidence, and uncertainty for each method. We will also clarify that the VLM is used during simulation distillation and is not queried by the deployed student policy. **[Insert final results and table reference.]**

Comment 5: The paper states that 22 objects were evaluated, while the current table shows only six representative objects. Please provide overall results for all 22 objects, for example separately for seen and unseen objects.

[Response:] We agree. The revision will report aggregate results for all 16 seen and 6 unseen objects and will provide per-object results for the complete 22-object evaluation in the supplementary material. Each aggregate will include the number of trials and a 95% confidence interval. **[Insert the final seen/unseen results and table references.]**

D Response to Reviewer #22

We sincerely thank the reviewer for the detailed and constructive comments. Below are our point-by-point responses.

Comment 1: The term “safety-critical” is too strong because the framework supplies empirical intervention rather than formal constraint satisfaction. State explicitly that safety violations remain possible.

[Response:] We agree. The revised manuscript will state explicitly that the method provides empirical risk reduction under task-specific event definitions, not a formal safety guarantee, and that nonzero violation probability remains in both simulation and physical deployment. We will revise the title, abstract, main text, and conclusion to remove or qualify claims that exceed this scope. **[Insert the final wording and section locations.]**

Comment 2: Demonstrate that the VLM is necessary by comparing against fixed thresholds, rule-based adaptation, Bayesian optimization, curriculum schedules, grid search, and a small learned threshold network.

[Response:] We agree that stronger conventional baselines are required. We will compare validation-selected fixed thresholds (using grid search and Bayesian optimization) with adaptive curriculum, rule-based feedback, learned-network, and VLM-guided controllers. The fixed methods will select one operating point, whereas the adaptive methods will update the operating point during training. All comparisons will use matched data and compute budgets where applicable and will report task success, unsafe-event rate, intervention rate, teacher queries, and training cost. **[Insert the final protocol, quantitative results, and table/figure references.]**

Comment 3: The Bellman critic predicts combined “safe-lift-success,” thereby conflating task success with safety. Separate safety risk and task value, and report calibration, false-negative rates, and reliability diagrams.

[Response:] We agree that the current joint target does not distinguish an unsafe outcome from a safe episode that fails to complete the lift. In the revision, we will **[state whether the critic is redesigned with separate safety and task heads, or justify and retain the joint target]**. We will report AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and the false-negative rate at the deployed threshold on held-out trajectories. **[Describe exactly how the revised critic enters the intervention gate and insert the results.]**

Comment 4: Teacher intervention is assumed safe, yet teacher policies may collide or fail. Quantify teacher unsafe rates and define how the framework responds when teacher and critic disagree.

[Response:] We agree that teacher takeover is not itself a safety guarantee. We will report success and unsafe-event rates for the teacher policies over the complete evaluation set rather than only representative objects. We will also specify how the controller handles disagreement between the critic’s assessment and the teacher action, including the fallback used when neither candidate action satisfies the acceptance rule. **[Insert teacher statistics, the fallback rule, and section/table references.]**

Comment 5: Provide exact VLM prompts, model version, decoding settings, image-selection procedure, output parsing, update frequency, latency, cost, and failure handling. Reproducibility may otherwise depend on an external proprietary service.

[Response:] We agree. The revised supplementary material will include all of the requested information, together with the permitted threshold range, smoothing rule, invalid-output handling, and deterministic fallback. We will archive the raw VLM outputs and threshold trajectories where allowed so that the adaptation process can be audited independently of the external service. **[Insert exact settings, measured latency/cost, archive location, and appendix reference.]**

Comment 6: Real-world unseen-object performance is weak: lift success falls to 60–66.7%, while unsafe episodes reach 40%. Claims of sufficient deployment robustness should be moderated.

[Response:] We agree. We will remove the claim that the current results establish sufficient robustness for deployment. Instead, we will present the physical experiments as an initial transfer evaluation and identify the unseen-object results as evidence of a remaining sim-to-real and contact-generalization gap. **[Insert the revised discussion and limitation-section locations.]**

Comment 7: Only fifteen real-world episodes per reported object produce coarse estimates. Increase repetitions and report confidence intervals and statistical comparisons across all 22 objects.

[Response:] We agree that the uncertainty of the current estimates must be made explicit. We will **[state the feasible number of additional trials]** and report per-object and aggregate seen/unseen results with 95% confidence intervals. Matched method comparisons will include effect sizes and appropriate statistical tests or bootstrap intervals. **[Insert the final trial count, analysis method, and results.]**

Comment 8: Safety-event thresholds for harmful collision, object escape, palm flipping, and workspace violation require physical justification. Include contact-force measurements and severity-weighted safety metrics.

[Response:] We agree that each event definition requires a physical and sensing rationale. The revision will list the numerical threshold, sensor or simulator signal, units, and justification for every event. In addition to binary episode incidence, we will report **[the available force, impulse, motor-current, penetration, or other severity proxy]**. If calibrated force measurements are not available, we will label the reported signal as a proxy and state this limitation explicitly. **[Define the severity-weighted metric and insert the section/table references.]**

Comment 9: Clarify whether VLM adaptation occurs only during simulation distillation or during physical deployment. If absent at deployment, the role of scene-level VLM reasoning should not be overstated.

[Response:] The VLM is queried only during simulation-based online distillation. During physical deployment, the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation. We will state this distinction in the abstract, method, experimental setup, and limitations, and will describe the VLM as a training-time adaptation mechanism rather than a deployment-time safety monitor. **[Insert the final section locations.]**

Comment 10: Compare training samples, teacher queries, interventions, computation, and final success across Vanilla DAgger, SafeDAgger, critic-based intervention, and VLM-guided intervention using multiple random seeds.

[Response:] We agree. We will add a unified comparison using the same training horizon and paired random seeds. For each method, we will report environment steps, student updates, teacher labels, teacher-controlled steps, intervention rate, compute time, VLM calls where applicable, final lift success, final unsafe rate, and unsafe-rate area under the training curve. **[Insert the number of seeds, quantitative results, and table/figure references.]**

We thank the Editor and the reviewers again for their thoughtful feedback. The requested revisions will help us present the method’s contribution more clearly, support its empirical claims more rigorously, and state its limitations more precisely.